MedXpert-CAD: A Multimodal Multi-Agentic System for Clinical Imaging Analysis via Model Context Protocol LLM-Driven Agentic Workflows

Mukhlis Raza[1], Saied Salem[1], Afnan Habib[2], Osamah Abdulmahmod[2], Hyunwook Kwon[2], Jamil Hussain[2], and Mugahed A. Al-antari[2(✉)]

[1] Department of Artificial Intelligence, College of AI Convergence, Sejong University, South Korea
[2] Department of Artificial Intelligence and Data Science, College of AI Convergence, Sejong University, South Korea
{raza.mukhlis, said.salem, afnanhabib, abdulmahmod.osamah, sad6601}@sju.ac.kr
{jamil, en.mualshz}@sejong.ac.kr

Abstract. Automating multimodal medical imaging is vital due to the complexity of integrating diagnostic processes and patient interfaces into cohesive clinical workflows. This paper presents MedXpert-CAD, a multimodal multi-agent system (MMAS) that employs large language models (LLMs) and vision encoders for an evidence-based VQA chatbot, supporting multi-task diagnosis for X-ray respiratory diseases and MRI lumbar spinal stenosis (LSS). MedXpert-CAD features four dedicated sub-agents: the supervisor agent, which queries users and directs them to modality-specific tools like X-ray respiratory disease analysis or LSS prediction; the online search agent, which retrieves data from PubMed or Google Search when needed; the X-ray expert agent, which processes images to provide multi-label classification, bounding box detection via saliency mapping, VQA responses, and structured reports; and the LSS expert agent, which handles MRI or DICOM files, offering sagittal and axial view segmentation, spinal pathological measurements, disk herniation diagnosis, and structured reports. MedXpert-CAD integrates the Model Context Protocol (MCP) with specialized agents to overcome AI model limitations, ensuring interoperable architecture and evidence-based diagnostics. The evaluation used the DeepEval framework and LLM-as-a-judge methodology, assessing task completion (TC), contextual relevance (CR), and tool accuracy (TN). GPT-4o showed exceptional performance with TC rates of 90% (X-ray) and 96% (MRI), and 100% TN in the single-agent framework, compared to Llama-3.2's 61% TC (X-ray) and 63% TC (MRI) with 100% TN. In the multi-agent framework, GPT-4o achieved 80% TC and 87% TN, surpassing Llama-3.2's 65% TC and 47% TN, with contextual relevance peaking at 83%. Segmentation efficacy peaked on sagittal views with a Dice Similarity Coefficient (DSC) of 96.43% and 99.66% accuracy, outperforming axial views (DSC 92.71%, accuracy 99.88%). Report generation metrics highlighted GPT-4o's superiority with METEOR 29%, ROUGE-L 19%, FrugalScore 57%, Blue-1 31%, and BERTScore-F1 84% for chest X-ray, and METEOR 18%, ROUGE-L 11%, FrugalScore 59%, Blue-1 11%, and BERTScore-F1 82% for

LSS. This system enhances patient care and empowers healthcare professionals with advanced imaging tools.

Keywords: Agentic AI; Large Language Model (LLM); Model Context Protocol (MCP), Medical Multimodal Integration; Reasoning-Action events; VQA AI chatbot.

1 Introduction

Large language models (LLMs) serve as fundamental components of the intelligent agents providing a complex reasoning capability towards actual-world scenario support [1]. The combination of zero and few-shot learning features in LLMs makes it possible to automate complex procedures which need detailed planning and multi-stage reasoning [2]. Healthcare organizations use conversational AI systems to improve operations by addressing experts' shortages and managing growing workloads. Like human intelligence, agentic AI integrates reasoning and actions to dynamically process multimodal external knowledge [3-5]. The GPT model is a multimodal AI system that enhances its processing capabilities by integrating the pair of image and text [6]. Agentic systems face difficulties when implementing multiple clinical decision-making (CDM) in a unique system to deal with various medical imaging modalities such as chest X-ray (CXR) and MRI scans [7]. Therefore, the effective techniques are needed to integrate knowledge of imaging benefits and risks, enabling the synchronization and management of complex medical data from CXRs and lumbar MRI 3D DICOM volumes [8]. Integrating various medical imaging data remains a significant challenge in developing a unified AI system that meets the needs of physicians, medical staff, and healthcare organizations for improved diagnosis and treatment planning. To address these challenges, the MedXpert-CAD system has been proposed and developed as a multimodal multi-agent system (MMAS) enhanced with Model Context Protocol (MCP) standardization for seamless medical image integration. MCP represents an open standard, where multiple AI applications can connect with various data sources through a universal protocol rather than requiring custom connectors for each integration. In healthcare applications, MCP enables AI systems to access real-time clinical data, external medical databases, and specialized tools through standardized interfaces, addressing the critical interoperability challenges that have hindered scalable AI deployment in clinical environments. Leveraging LLMs and vision encoders, the MCP-enhanced MMAS platform powers an evidence-based VQA support chatbot, forming the foundation of MedXpert-CAD. Our main contributions are:

1. MedXpert-CAD, a novel explainable multi-agent chatbot MCP-based CAD system, analyzes multiple medical image modalities with transparent, evidence-based reasoning. Four autonomous sub-agents interact synchronously to improve functionality, integration, and doctor trust in AI-driven diagnostics.
2. An innovative deep search feature integrates with an online search agent accessing PubMed and Google Scholar, as well as a knowledge base (Vector and Graph databases), enabling context-aware diagnoses by retrieving accurate medical knowledge for image analysis and text reporting.

3. The custom MCP server for chest X-ray expert agent provides advanced analysis through multi-label disease classification, abnormality localization, saliency maps for visual explanations, and structured radiology reports, surpassing conventional classification methods.
4. The custom MCP server for LSS expert agent enhances lumbar MRI assessment with advanced analytics, including axial and sagittal segmentations, disk herniation diagnosis, and quantitative pathological measurements to improve complex spinal health conditions.

2 Related Works

The recent advancements in tool-augmented LLMs demonstrate substantial improvements which advance their practical operation capabilities for better performance and efficiency [9]. The recent multiple advances converge to enhance LLMs' ability to use the diagnostics tools correctly through specific agents for multiple tasks generalization and efficacy purposes. Parisi et al. (2022) [10] demonstrate knowledge-intensive reasoning using a text-only augmentation mechanism combined with iterative self-play methods. In 2024, Chen et al. [11] built an inference trajectory optimization framework based on impractical decision tree exploration to generate better performance and generalization capability. Healthcare implementations of tool-augmented LLMs create valuable opportunities aiming to solve medical problem complexities with multimodal image and non-image sensitive datasets. Li et al. (2024) [12] proposed their MMedAgent system to handle various medical operations with some tools to process different medical data modalities. Shen et al. (2024) [13] showed how MATMCD represents a multi-agent system which applies integrated multimodal data to perform knowledge-driven causal discovery within healthcare settings. According to Yuan et al. (2024) [14], the medical application of LLMs shows promising agentic AI systems where personalised models and complex clinical decision-making based on LLM capabilities in medical science is crucial. The processing ability of multimodal LLMs for medical information depends heavily on ongoing optimization and ethical oversight. None of the frameworks provides a unified Model Context Protocol (MCP) that allows clinical real time image processing, database queries and reports generation; MedXpert-CAD addresses this need with elastic MCP servers.

3 Methodology

MedXpert-CAD is a multi-modality multi-agent system (MMAS) that is a tool tailored to analyze clinical images and non-image data, incorporating cutting-edge diagnostic facilities to handle the complexities of medicine represented in Fig. 1. The system is characterized by special AI agents, with deep searches and evidence-based logical reasoning, guaranteeing that intensive medical work does not interfere. The architecture has three different MCP servers namely: Deep Search MCP Server that retrieves literature in terms of PubMed and Google Search, the Chest X-ray MCP Server that classifies them, creates a report, visualizes the heatmap, and detects the bounding box, and

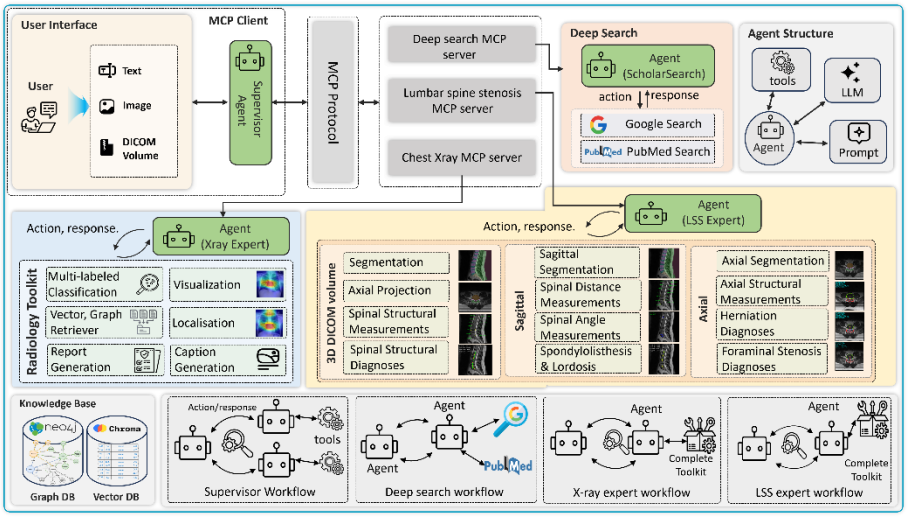


Fig. 1. MedXpert-CAD Multi-Agent Architecture: Workflow of the Four-Step Clinical Imaging Analysis Pipeline (User Input → Supervisor Routing → Tool Execution → Aggregated Output)

the Lumbar Spine Stenosis MCP Server that retrieves MRI vector database, processes DICOM, and detects the stenosis. With the help of the MCP Client and MCP Protocol, these servers connect with the Supervisor Agent, X-ray Expert Agent, LSS Expert Agent, and Deep Search Agent. The workflows involve knowledge sources, such as PubMed and Google Scholar, as well as agent tools and LLMs, which allow a robust and extensible basis of the diagnostic process and include state-of-the-art workflows, such as the work of X-ray experts, the work of LSS experts, and deep search operations.

3.1 Overall Systematic Architecture

The hierarchical structure of MedXpert-CAD includes four dedicated agents along with the three custom MCP servers, which are designed to improve clinical imaging workflows. Acting as the main interface of the system the supervisor agent executes the specified MCP server according to the user's question. The X-ray expert Agent analyzes frontal and/or lateral X-ray image views from MIMIC-CXR [15] using our radiology toolkit to provide multiple services: multi-label classifications, explainable Grad-CAM saliency maps to localize the pathological affected areas on the entire images, and a structured medical diagnostic report for comprehensive and rich diagnostic and treatment plans. The LSS expert Agent specializes in lumbar spinal MRI [16] analysis to provide automatic and accurate sagittal and axial image segmentations, related pathological measurements, and generated a comprehensive structured medical diagnostic report. Our LSS toolkit could handle the 3D DICOM volume in addition to a single axial or sagittal images towards the quantitative analyses of stenosis grading, disc

herniation, lumbar lordosis, and spondylolisthesis. Data splitting technique of 70% training, 20% validation and 10% testing was used for both modalities, with pre-processing by resizing, augmentation and normalization prior to analysis.

3.2 Action-Observation-Thought (AOT)

MedXpert-CAD deploys sub-agents to complete AOT [4] cycles by interacting with the MCP-standardised tool interfaces by employing reasoning to undertake action steps, and gathers the tool observation to advance the thoughts. In contrast to pure chain-of-thought reasoning which uses only text generation as a connective element, our AOT framework incorporates external medical tool execution used in the process of reasoning. The *Action*→*Observation*→*Thought* cycle leads to the dynamic use of tools implemented in MCPs (DICOM processing, vector database query and retrieval query, classification models), as well as observing the structured medical output, and integrating it into the next iteration of processing. MCP integration makes it possible to have standardized tool invocation such that agents interact with special medical capabilities using uniform protocols. When the agent parses the lumbar MRI data, it invokes DICOMProcessor tool, views the extracted metadata and segmented labellings, examines the specified artifacts of concrete medical evidence and utilizes it to conduct diagnostic reasoning, without depending solely on parametric knowledge.

3.3 Evaluation Criteria and Executional Learning Environment

The evaluation framework employs *DeepEval* the unit test uses LLM as a judge consists of three essential performance metrics for complete agentic system assessment as summarized in Eqs. (1-3). First, *TaskCompletion* (TC) determines the agent's accomplishment tasks. Second, the *ContextualRelevancy* (CR) checks the correctness of the generated action responses of the available contexts to achieve relevant and coherent output. Third, *ToolCorrectness* (TN) metric that assesses LLM agent's function/tool calling ability. This set of metrics combines delivering comprehensive multi-dimensional assessments between goal achievement, proper context placement, and delivering accurate information.

$$\mathrm{TC} = AlignmentScore(Task, Outcome) \tag{1}$$

$$\mathrm{CR} = \frac{Number\ of\ Relevant\ Statements}{Total\ Number\ of\ Statements} \tag{2}$$

$$\mathrm{TN} = \frac{Number\ of\ Correctly\ Used\ Tools}{Total\ Number\ of\ Tools\ Called} \tag{3}$$

The experiments are conducted on a PC equipped with an i7-1400K processor, 64 GB of memory, and an Nvidia RTX 4070 Ti Super GPU.

4 Results

The evaluation process of MedXpert-CAD is performed on both benchmark public medical datasets of chest X-ray and lumbar spine MRI scans [16]. The research examines single-agent and multi-agent frameworks by using two LLMs: GPT-4o [17] and Llama-vision3.2 [18]. The results are summarized in Table 1. GPT-4o achieves Task Completion (TC) scores of 80% for all agents' scenario for CXR and MRI tasks, respectively. In comparison, Llama-Vision 3.2 attains TC scores of 65% for CXR and MRI interpretations. GPT-4o exhibits higher tool correctness than Llama-Vision 3.2, with CXR and MRI recorded at 87.6%, respectively. Additionally, GPT-4o outperforms Llama-Vision 3.2 in Contextual Relevancy (CR), scoring 83% for CXR and for MRI. Meanwhile, Llama-Vision 3.2 shows a CXR and MRI tool correctness of 47% with CR scores of 68% for CXR and MRI. Overall, GPT-4o appears to benefit from superior training data selection and model optimization strategies, likely due to its proprietary advancements. The minimal fluctuation across different modalities suggests that multi-agent integration may suffer from increased complexity or error accumulation during reasoning-action processes. Our multi-label classification tool leverages the ensemble capabilities of DenseNet [19] and ResNet [20] to effectively diagnose 14 respiratory diseases: Atelectasis, Cardiomegaly, Consolidation, Edema, Enlarged Cardiomediastinum, Fracture, Lung Lesion, Lung Opacity, No Finding, Pleural Effusion, Pleural Other, Pneumonia, Pneumothorax, Support Devices. For both LSS sagittal and axial segmentation tools, we employ ensemble hybrid approaches combining CNNs and transformers using the random search automatic hyperparameter optimization approach [21] For sagittal images, the top three selected models are U-Net++, U-Net, and SwinUNETR, while for axial images the top models are SwinUNETR, Attention U-Net, and U-Net ++. Additionally, extensive experiments have been conducted to select, optimize, and develop single CXR and MRI tools, ensuring the refinement and finalization of their AI structures. Fig. 2 shows the summary of CXR multi-label classification results as well as the segmentation performance of LSS axial and sagittal images.

Table 1. Performance comparison of MedXpert-CAD for single-agent against multi-agent frameworks using GPT-4o and Llama3.2 LLMs across X-ray and MRI Tasks.

Framework	Model	Modality Agent	TC (%)	CR (%)	TN (%)
Single-Agent	GPT-4o	X-ray Expert Agent	90	95	100
		MRI Expert Agent	96	59	100
	Llama-3.2-11B-Vision	X-ray Expert Agent	61	66	100
		MRI Expert Agent	63	47	100
Multi-Agents	GPT-4o	All Agents	80	83	87
	Llama-3.2-11B-Vision	All Agents	65	68	47

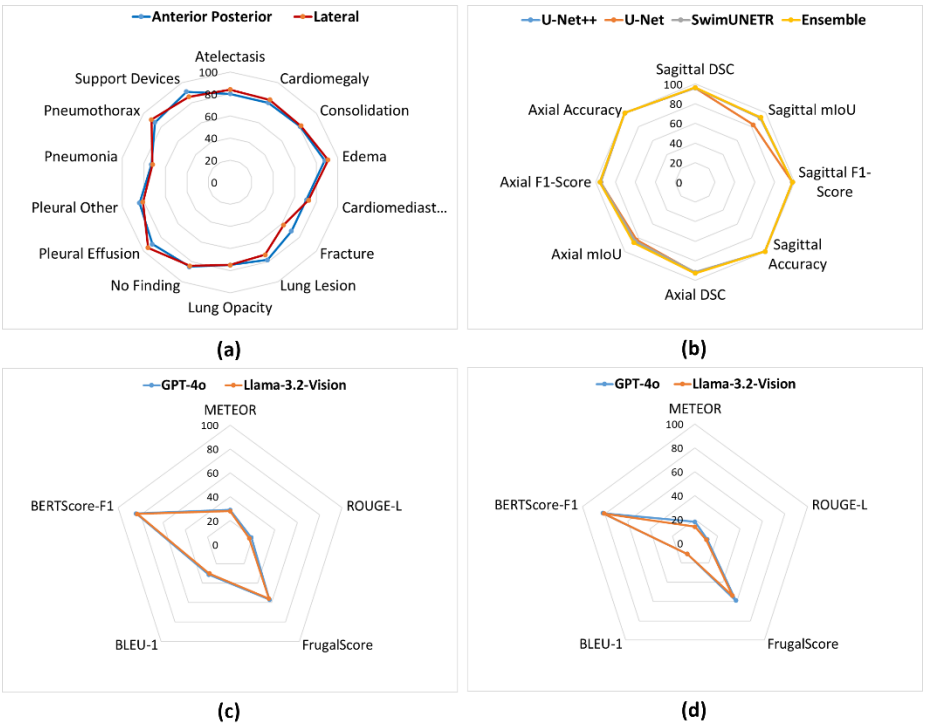


Fig. 2. Performance evaluation of the proposed toolkits on the MedXpert-CAD system: (a) multi-class classification of the CXR tool for Anterior-Posterior (AP) and Lateral image views, (b) multi-class segmentation for sagittal and axial tools, and (c)-(d) Medical text report generation results for CXR and LSS agents, respectively.

As shown in Fig. 2, GPT-4 produces better accuracy, and coherence reports for X-ray images than Llama-3.2-Vision. For LSS tasks, GPT-4o demonstrates a slight advantage over Llama-3.2-Vision. MedXpert-CAD demonstrates its capability to use state-of-the-art LLMs effectively in handling multiple medical imaging simultaneously.

5 Discussion

MedXpert-CAD serves an end-to-end chatbot platform, with the ability to have smoother diagnostic workflow, with the capability to process 2D images and 3D DICOM volumes, including upload, classification, segmentation, and measurement, as well as forecast, prediction, and structured reporting (Fig. 2). The Model Context Protocol (MCP) has been integrated to normalize the tool interface allowing efficient

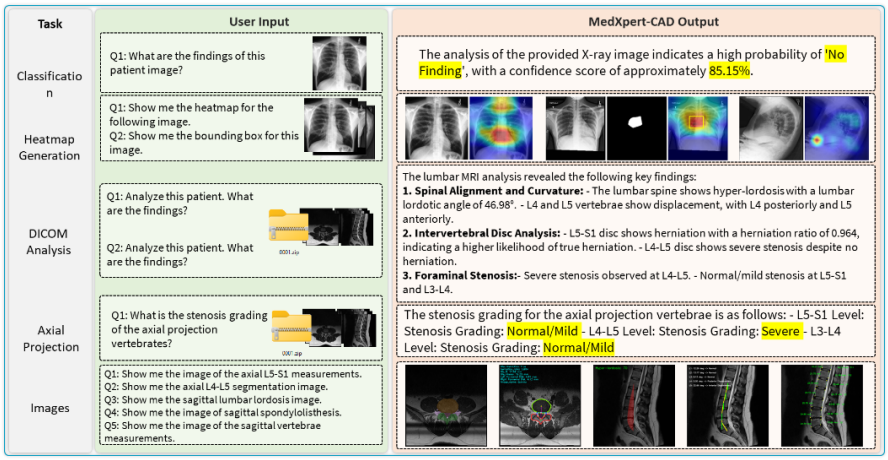


Fig. 3. Example Input-Output Workflow of MedXpert-CAD: User Queries and System Responses for X-ray and MRI Analysis (Thoughts, Actions, and Values).

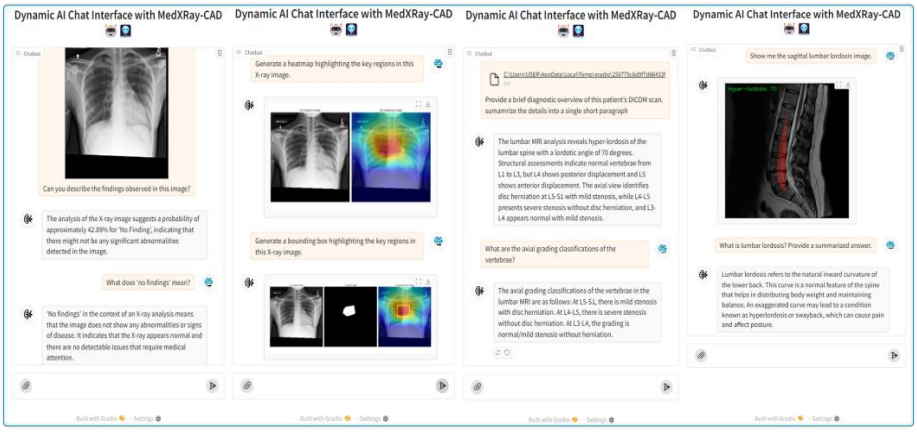


Fig. 4. Chatbot illustration, how users can interact and the chatbot response.

coordination among a proficient agent specializing in real time image processing or database queries. Llama-3.2 test results indicate that it is an equivalent replacement of GPT-4o when considering open-source evaluation of finances, and it does not perform worse. Fig. 2(a) provides the results on the classification of the chest X-ray using the analysis of the AUC curves, showing the high robustness of diagnostic findings at all modalities. Figure 2(b) demonstrates segmentation model performance, our ensemble model performed better than single methods, especially in sagittal and axial slices. Figs. 2(c) and 2(d) show the report generation results, where (c) implies better quality of the

chest X-rays report (e.g., improved METEOR and BERTScore-F1) and (d) represents the competitive results of LSS report generation. To support the robust tool invocation and evidence-based outputs, we designed three MCP servers, including Chest X-ray, Lumbar Spine Stenosis, and Deep Search, as illustrated in Figs. 3 and 4, which indicated dynamic interactions with users. Multimodal data is processed by the chatbot, which provides dependable results within 5-9 seconds, including tool interactions and analyses. MIMIC-CXR and Lumbar Spine MRI datasets were utilized; however, broader multi-institutional validation remains necessary. Inference is completed within a few seconds indicating potential for real-time application. System-level, modular agent integration constitutes the primary innovation, rather than novel technical modules. Comparative analysis with frameworks such as AutoGen, DyLAN and MDAgent is planned for future work extension.

6 Conclusions

MedXpert-CAD represents a significant advancement in AI-driven clinical imaging analysis, introducing a multimodal multi-agent system (MMAS) that integrates LLMs, vision encoders, and domain-specific tools to streamline complex diagnostic workflows. Key innovations include Modular Architecture, Evidence-Based Diagnostics, and Interpretability. While the framework excels in static imaging modalities (X-ray, MRI), limitations include its current exclusion of dynamic modalities like ultrasound and dependency on continual fine-tuning for emerging datasets. By combining open-source adaptability with clinician-centric design, MedXpert-CAD sets a new standard for scalable, trustworthy AI in healthcare, empowering medical professionals with advanced, interpretable diagnostic tools.

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Disclosure of Interests. Not applicable.

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